

A Proposal for Automatic Evaluation of Human Functional Limitations in Activities of Daily Living

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Abstract—This work presents a novel proposal aimed at automating the assessment of human functional limitations. It represents an interdisciplinary effort to develop and clinically evaluate an automated system for objective functional assessment, and is being carried out within the EYEFUL research project. Functional evaluation is a complex task that requires a holistic approach, considering various components like cognition, motor skills, and executive functions. The EYEFUL project has created an intelligent space within a fully equipped apartment, which includes the deployment of advanced technology (cameras, microphones, and wearable devices), to capture data from subjects undergoing assessment. In this paper, we provide an extensive overview of the project's architecture, including the selection of Activities of Daily Living (ADLs), the recruitment of subjects, the validation methodology, and the experimental setup. We describe the specialized apartment and sensor hardware deployed for data collection, along with the initial machine learning systems for

data analysis. Additionally, we analyze the existing state-of-the-art research in automatic functional performance evaluation and present our proposal. This encompasses ADL selection, database design, recording considerations, and an overview of technological modules to be implemented. We also discuss the current status of select modules, including quantitative and qualitative results and references to ongoing work. In conclusion, the EYEFUL project is pioneering automated functional assessment, integrating advanced technology and interdisciplinary collaboration. It aims to provide healthcare professionals with accurate, comprehensive, and clinically validated evaluations of human functionality during ADLs, addressing the complex nature of functional limitations.

Index Terms—Automatic human functional evaluation, intelligent spaces, activities of daily living, occupational therapy, machine learning, multisensorial system.

Received 30 September 2023; revised 29 July 2024 and 11 November 2024; accepted 14 December 2024. Date of publication 24 December 2024; date of current version 8 December 2025. This work was supported in part by the Spanish Ministry of Science and Innovation MICINN/AEI/10.13039/501100011033 under Project EYEFUL-UAH (PID2020-113118RB-C31), Project EYEFUL-URJC (PID2020-113118RB-C33), and Project ATHENA (PID2020-115995RB-I00), in part by the Spanish Ministry of Education and Professional Training (S.F.L. with Beca-Colaboracion 2022-2023), by CAM under Project CONCORDIA (CM/JIN/2021-015), and in part by INVESTIGO Program. The project has been approved by the Research Ethics Committee of the Universidad Rey Juan Carlos, with reference code 2202202202204922. This study has also been registered in ClinicalTrials.gov, with identifier NCT05385328. (Corresponding author: Javier Macias-Guarasa.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Research Ethics Committee of Universidad Rey Juan Carlos under Application No. 2202202202204922, and performed in line with the regulations of the URJC Research Ethics Committee.

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Digital Object Identifier 10.1109/JBHI.2024.3521931

I. INTRODUCTION

THIS work presents a novel proposal aimed at automating the evaluation of human functional limitations. This initiative is part of the EYEFUL project, which represents a significant interdisciplinary effort and marks a pioneering step towards the development and clinical evaluation of an automated system for the objective assessment of human functional limitations.

Evaluating functionality is a complex task, requiring a holistic approach that integrates tools from various domains. Additionally, it requires considering contextual factors, as highlighted in previous research [1], with a growing recognition of the importance of assessing diverse components of health, including cognition, motor skills, and executive functions like planning and task execution, within intelligent environments that mimic real-world settings [2].

To achieve its ambitious goals, the EYEFUL project has undertaken the task of creating a realistic adapted apartment, tailored for functional assessments. This environment is equipped with advanced technology, including cameras, microphones, wearable devices, computing nodes and communication devices that collectively capture and process data from subjects undergoing assessment. This dataset is pivotal for developing and rigorously testing fusion algorithms and data analysis methodologies. Collaborating with occupational therapists (OTs) ensures that

the algorithms provide insights directly relevant to functional assessments.

In this paper, we provide an extensive overview of the EYE-FUL project's architecture. We delve into the selection of Activities of Daily Living (ADLs) by OTs for assessment, the meticulous procedures for subject recruitment to construct a specialized dataset, the methodology underpinning human functional validation, and a detailed description of the experimental setup. We also explore the specifically adapted apartment and the sensor hardware deployed for data collection. To facilitate automated functional assessments, we describe the main software systems designed for data analysis.

This study makes several contributions to the field of automatic human functional evaluation, leveraging advanced technologies within intelligent environments: (i) A rigorous approach to carry out studies aimed at evaluating human functional limitations, highlighting the required interdisciplinary approach by uniquely combining expertise from OTs and engineering teams exploiting state-of-the-art machine learning techniques; (ii) a carefully designed and realistic experimental environment, in which the design of the sensorization, computing and communications equipment is fully guided by user and clinical considerations; (iii) a user and clinically guided proposal for the database to be recorded and labeled (including both sensor data and clinical data supported by well established scales); (iv) a proposal and initial implementation of a system architecture leveraging the required state of the art technologies to efficiently estimate the required metrics and inferences with results that, despite being preliminary and requiring further integration, already shows promising improvements over existing methods. We believe that our proposal will help other research groups in similar tasks in the area of automatic human functional evaluation, that has received little attention in the literature with the holistic approach that we consider.

Section II includes a critical analysis of the existing state-of-the-art research landscape in automatic evaluation of functional performance. Section III fully describes our methodology and proposal. Section IV addresses the current status of some of the technological modules (including quantitative and qualitative results), and references to current work. Finally, Section V summarizes the main conclusions and future work.

II. PREVIOUS WORK

According to the World Health Organization (WHO)'s International Classification of Functioning, Disability and Health (ICF) framework, *disability* is a multidimensional concept that takes into account an individual's health condition, limitations in performing activities, and restrictions in participating in various life situations. It emphasizes the interaction between the person and the environment, giving that disability is not solely a result of a person's health condition but it is also influenced by external factors [3].

The main indicator of functionality is the performance of Activities of Daily Living (ADLs). Functional impairment can be triggered by many diseases, which negatively affect the ability to independently perform basic ADLs by altering several factors, including physical and cognitive factors [4].

The evaluation of functional performance can be approached using various methods, including direct observation, self-administered tools [2], or family interviews [5]. In the context of assessing dependency in Spain, the evaluation relies on interviews regarding functional capacity in ADLs and unstructured observations [6]. Notable examples of functional assessment tools include the Barthel index [7], and the Katz index [8]. However, it is important to note that no universally recognized "Gold standard" has been established for functionality assessment [1], [7], and many of these methods may not possess the necessary sensitivity to accurately determine an individual's true level of functionality [9].

Previous studies have noted that existing instruments for assessing disability have presented limitations in practice, as the information they detail is not related to the quality of execution [5], requiring the investment of a long period [1], [2]. Furthermore, these instruments implicitly produce biases because they are subjective [8], [9], gender and culture biases [10], evaluator biases [11] and interviewee biases [1], [2]. Interviews also require a reliable source of information [12], [13]. In the case of self-administered tools, they are subject to recall bias, since, based on retrospective recall of performance [4], individuals may consider the performance problem not important enough to express, be aware of but deny it [5] or not have the ability to accurately and/or realistically report it [7], [11], [13], as it is common for people to consider that they can perform the activity even though they do not [5]. Observational instruments may be negatively affected by observers' interventions or biases [9] and the demand for training of professionals to administer them [5], [9], such as the "Assessment of Motor and Process Skills" (AMPS), which does not indicate the degree of independence or assistance required in the activities that the person performs [12].

Also, no techniques are used that accurately analyze the actual characteristics of the disability, which focus directly on the biological and physical patterns, objectives of the impairment [8]. Technology can bring improvements to the evaluation of functional performance [4], through the monitoring of activities and the real-life behaviors demanded by them, it can allow the development of more effective evaluations [9], [14] when objectively quantified [9], and even contribute to the early detection of changes in functionality [4]. Intelligent devices through environmental sensors favor the automatic recording of information, by the monitoring of ADLs, being a valid method to characterize the parameters of activities [15] such as frequency, intensity, or duration [14] and behavior patterns including gestures, movement, location, interactions, actions [4] and motor behavior [14]. On the other hand, it can be a more representative measure of a person's real capacity when carried out in real environments [4], [11], [15], so the resulting data are ecologically valid, eliminating biases [13].

Besides, automatic and continuous recording can provide more information, helping professionals to better understand people's needs, situations, and diseases [4], as well as facilitating decision making to maintain or improve the health status of the person or caregiver, in which their deterioration in the level of independence and functionality of daily activities is impacting their health and quality of life [16].

As reported in [17], different studies have analyzed the relationship between activity performance, collected using technological sensors and clinical evaluation tests. These works focused on motor parameters, such as falls, fragility, walking profiles, motor performance [10]; cognitive elements [9], [13], [14]; or both, as in [15], that analyzed walking and neuropsychological state. Also, psychological parameters, such as emotional state [11], and time-related variables, such as changes in routines and schedules of activities [14] have been evaluated. The execution of instrumental activities has been analyzed as a predictor of functional capacity in [11]; also in [10] authors evaluated the functioning in activities; however, they presents limitations such as not correlating the results with other standardized tools. In [18] authors developed sensors that predict functionality by equating it to the AMPS score [1], evaluating whether the performance of a set of instrumental activities can detect mild cognitive impairment, but did not consider variability in the performance of different individuals who performed them. It has been shown that intelligent systems can reduce the socioeconomic and social impact of people with disabilities [14], as they are useful for evaluating the health status of the person and detecting the appearance or progression of some diseases and/or disorders associated with functional impairment [15].

However as explained in [19], the diagnostic value of these systems has not been widely recognized in research [13], as most of the current technological methods of functional assessment focus only on one aspect of functionality [4], especially in the cognitive field [13]. Moreover, as detailed above, these methods have some limitations because functionality assessment requires a holistic approach and the integration of several tools that include more than one domain, as well as contextual factors [4]. It has also been there indicated the need to assess the components and determinants of health such as cognition, motor skills, or executive functions; and also such planning and task fulfillment in intelligent environments that approximate ecological environments [2].

III. A PROPOSAL FOR THE EVALUATION OF FUNCTIONAL LIMITATIONS

In this section we describe the overall design of our proposal for the evaluation of human functional limitations, starting with the general user-centric design methodology, followed by the strategy we applied to decide on the selection of ADLs. We then describe the experimental environment in which the evaluation will be carried out, including the deployment of sensors, computing equipment and communication devices. The description of the database generation and labeling processes is discussed next, and we finish with an overview of the technological systems architecture that will fulfill the requirements in terms on inference and metrics generated.

A. User-Centric Design Methodology

The complexity of the functionality assessment task being tackled in the EYEFUL project requires a holistic approach and the integration of several tools that include more than one domain

(with OTs working side by side with engineers), as well as contextual factors and ADLs definition in intelligent environments. In such context, an ad hoc User-Centric Design [20] (UCD) has been adopted.

UCD is an iterative design process in which designers focus on the users and their needs in each phase of the design process, via a variety of research and design techniques, to create highly usable and accessible products for them. In our case, the close interdisciplinary collaboration between the OTs and the engineers guarantee that the users are involved along the full process.

The first work-package of the EYEFUL project is specifically in charge of the application of such user-centric methodology, where we defined four tasks dealing with the four stages in the standard UCD approach, as follows:

- 1) For context inclusion: definition of ADLs to be assessed, driven by the OTs with the direct patient's needs knowledge. In this process, the opinion and real experience of patients from different entities that collaborate and support the project were directly considered. This direct reflection by the patients allows for considering also a patient-centered design, which takes into account the real limitations faced by these people in their ADLs. The process of collecting patients' opinions was carried out by OTs, providing their knowledge of the patient's functional limitations and needs (see section III-B).
- 2) For requirements analysis: design and deployment of the sensorization configurations, taking into account the required ecological approximation (see section III-C).
- 3) For development: extraction of quantitative metrics and their relationship with qualitative measurable functional parameters, in a joint work of engineers and OTs in order to consider the technical constraints and the users' needs (see section III-F).
- 4) For evaluation: design and application of the technical and clinical validation processes, driven again by the OTs that are always next to the patients when performing each evaluation, and are aware of the rigorous procedures to establish the clinical validity of the developed systems (see sections III-D and III-E). Furthermore, the subjects' opinions and feedback were also captured by means of questionnaires that were administered before and after the actual recordings for each subject.

Therefore, with our proposed methodology, UCD is performed through its fundamental aspects, in which the needs and interests of the users are being fully considered.

B. Selection of Activities of Daily Living

The selection of appropriate Activities of Daily Living (ADLs) for the evaluation process was a crucial step in our study. This process involved a meticulous four phases procedure aimed at ensuring the comprehensiveness, objectivity, and universality of the chosen activities.

In the first phase, a panel of occupational therapy experts designed the initial list of ADLs. These experts considered the classification hierarchy of activities provided by the Assessment

of Motor and Process Skills (AMPS, [21]), a well-established tool for assessing functional capacity. To include the patients perspective, the group of occupational therapy experts from the Universidad Rey Juan Carlos (URJC) worked closely with volunteer patients from the different entities with which the project had a collaboration agreement, to define which of all the existing ADLs posed a greater challenge or effort for them to be able to carry them out. Our experts assessed each activity based on several criteria, including physical effort, safety, efficiency, and the need for assistance.

Furthermore, to enhance quality and objectivity, the experts evaluated the difficulty level of each ADL, focusing on activities requiring various forms of coordination, such as general coordination movements, fine motor skills, hand-eye coordination, and attention.

The initial list of selected activities is as follows: 1) Sweeping the floor of a room, 2) setting the table for four people, 3) putting on socks and shoes without laces, 4) brushing teeth, 5) making a sandwich, and 6) Prepare two hot drinks.

In a second phase, the group of experts in occupational therapy of the University Rey Juan Carlos (URJC) described all the ADLs with the highest level of detail possible, taking into account the characteristics of movement and cognitive processing of the actions. The executions were broken down according to the parameters of presence, space, speed and sequence, while the person was performing an activity. The relevant objects required, the starting and ending position, the actions and sequences, the actions required, and the performance instructions were established.

In the third phase, the work developed up to that moment was sent to a group of experts in clinical work so that they could elaborate a proposal selecting three activities out of the original list, with the criteria of providing enough information to detect limitations in the patients' functionality: 1) Sweeping the floor of a room, 2) setting the table for four people, and 3) putting on socks and shoes without laces.

This list was also subjected to a fourth phase of analysis by physicians experts in dependence conditions, who finally validated the suitability of these three activities to observe the elements involved in the evaluation of dependence.

To finally evaluate the feasibility of the proposal, a joint work between the OTs and the engineering group was carried out to first define the required inferences to be generated by the automated system, and the observable related functional parameters. The final interdisciplinary task was designing the research and development efforts oriented to the design of the adequate technological modules to infer the expected quantitative and qualitative results, which is actually an ongoing work being progressively refined.

C. Description of the Experimental Environment

The physical deployment of the proposal was done in a realistic fully equipped apartment, completely tailored for functional assessment tasks and adapted for persons with disabilities. It is located in the URJC campus, distributed as shown in Fig. 1. The apartment includes a living room, a kitchen, a bathroom and a

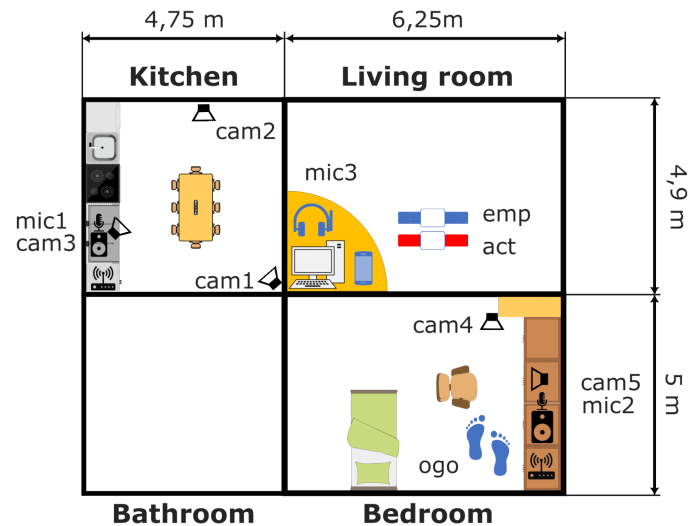


Fig. 1. URJC apartment floor plan.

bedroom. The selected ADLs are carried out in the kitchen and the bedroom.

The technological equipment deployed in the apartment is designed to capture the data streams that are relevant for the task, namely (we provide the brand and model, and the identifier of all of them in Fig. 1):

- Sensor equipment:
 - Two Stereolabs Zed 2i [22] RGBD cameras, that capture color (RGB) video and depth (D) streams: *cam1*, and *cam4*.
 - Three Intel Realsense D455 [23] RGBD cameras: *cam2*, *cam3* and *cam5*.
 - An Empatica E4 [24] Medical wristband, that captures physiological parameters through four sensors (photoplethysmography sensor, electrodermal activity sensor, a 3-axis accelerometer and an optical thermometer): *emp* (located in the dominant hand).
 - An ActiGraph wGT3X-BT [25] Medical wristband, with a tri-axial accelerometer: *act* (located in the non-dominant hand).
 - A pair of Moticon OpenGo Insoles [26], that captures plantar pressure in 16 sensors and includes a 6-axis Inertial Measurement Unit (IMU): *ogo*.
 - A far-field EPOS SP 30 [27] microphones with speaker, to capture spoken interactions and provide spoken instructions: *mic1* and *mic2*.
 - A MAIRDI M890BT Wireless Headset [28], to allow the operator to interact with the user inside the rooms while carrying out the ADLs.
- Computing and communication equipment:
 - A PC workstation that controls cameras *cam1*, *cam2* and *cam4*, and runs the overall system to control and synchronize the full recording procedure.
 - One NUC PC system that controls camera *cam3* and the EPOS device in the kitchen.
 - One NUC PC system that controls camera *cam5* and the EPOS device in the bedroom.

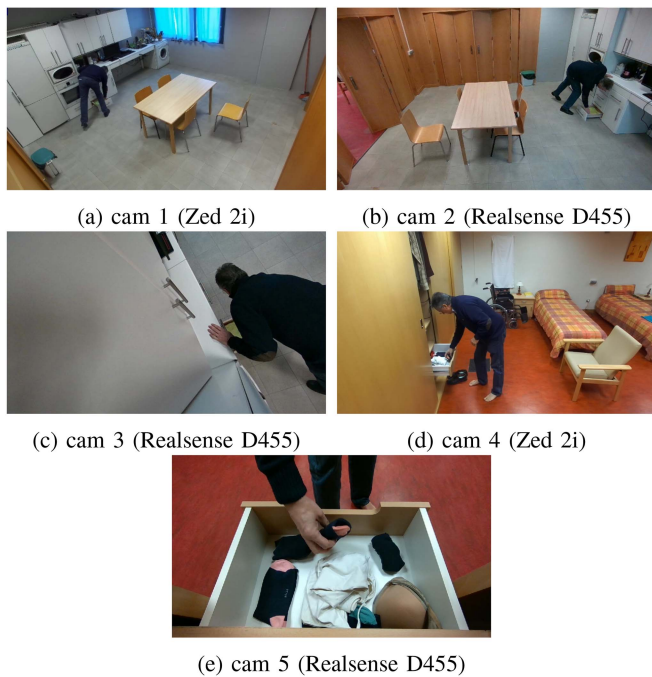


Fig. 2. Point of view of cameras in the recording environment.

- One mobile phone to control the insoles and the Empatica E4 wristband.
- One TP-LINK AX55 wireless router to allow fast communication among devices.

The kitchen is equipped with 3 cameras (cam1 and cam2 to provide a general view of the room, and cam3 to provide a top detailed view of the cupboard and drawers area), one far-field microphone+speaker EPOS SP 30 device, and one NUC PC system. The bedroom is equipped with 2 cameras (cam4 to provide a general view of the room, and cam5 to provide a top detailed view of the drawers area inside the cupboard), one far-field microphone+speaker EPOS SP 30 device, and one NUC PC system. Fig. 2 shows an example of the cameras' views in every room. The frames are synchronized for kitchen cameras (1, 2 and 3), and for bedroom ones (4 and 5).

Sensor placement criteria has been carefully considered in our experimental environment. Regarding the smart wristbands, our recording protocol dictate that they must be securely fastened so that they fit tightly around the wrists. The E4 wristband is fitted in the dominant hand and the ActiGraph in the non-dominant hand, to provide consistency to the acquired data. The correct placement of the insoles is guaranteed as their sizes correspond to the usual size of the users. Cameras have been carefully placed to maximize the viewing angle, ensuring full view of the person and relevant scene objects, without occlusions from at least one of them at each instant.

Controlled placement is initially considered as it eases calibration and further analysis. When considering applications in less controlled environments, the same general criteria would apply. Some of the technological modules would be relatively independent of camera placement, provided an initial calibration process is carried out to allow for sensor data fusion.

Nevertheless, we will devote future work to quantify the sensitivity of the results to sensor presence and placement.

The PC workstation running the control system is located in the living room, where you can also find a mobile phone in charge of controlling the insoles and the E4 wristband.

Communication among devices is done by either a WiFi connection (among computing nodes) or a Bluetooth one (between the blue color elements and the computing nodes).

D. Database Generation

The design, acquisition and labeling of a realistic dataset in the target application domain is a crucial task in EYEFUL.

Data will be collected for 90 subjects, equally divided in two cohorts of subjects with and without functional limitations (a recognized degree of dependence), so that a thorough clinical validation will be carried out.

The inclusion criteria for the subjects of the two cohorts include being over 18 years of age and signing the informed consent to use the collected data approved by the Ethics Committee before participating in the study. The exclusion criteria also state that subjects with the presence of cognitive, physical or sensory deficit that impedes understanding the questionnaires or carrying out the tests will also be excluded.

Healthy subjects need to present a score higher than 108 on the Functional Independence Scale (FIM) [29] and/or a score lower than 25 on the Barthel index [7]. For the subjects with recognized dependence, they need to have an official recognized degree of dependence by the Government, or a Barthel index score between 25 and 75. In any case, they also need to have the ability to move around independently, with or without assistive elements (canes, crutches, walkers, etc.).

Among the healthy subjects, the distribution between sexes is roughly equal (53.4% female and 46.6% male), with an average age of 49 years (17.7% older than 70, and 82.3% between 18 and 69). Regarding the subjects with dependency recruited so far, their distribution between sexes is also roughly equal (51% female and 49% male), with an average age of 55 years (14.3% older than 70, and 85.7% between 18 and 69).

All the subjects carry out the three selected ADLs with all the sensor equipment properly working. In addition to that, they are requested to fill in a set of questionnaires related to widely accepted scales, to finally allow assessing the validity of the scores provided by the EYEFUL evaluation:

- Clinically Useful Anxiety Outcome Scale (CUXOS-D) [30]: a brief item self-report measure designed to assess the severity of anxiety symptoms in adults with a diagnosed anxiety disorder or depression.
- Assessment of motor an process skills (AMPS) [31]: an observational assessment that measures the performance quality of daily living activities in a natural environment.
- Montreal Cognitive Assessment (MoCA) [32]: a brief test that helps health care professionals quickly determine whether a person has abnormal cognitive function.
- The Allen Cognitive Level Screen (ACLS) [33]: a standardized, evidence-based screen of functional cognition.

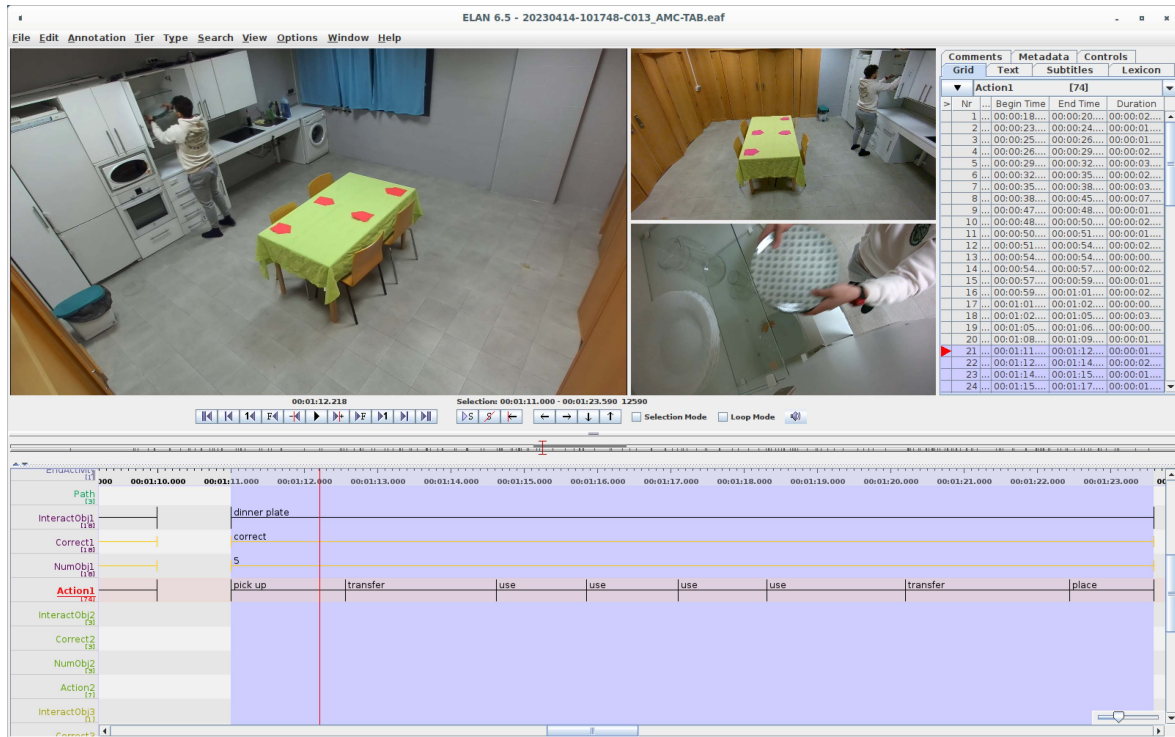


Fig. 3. An example of a labeling session with the ELAN application.

- Six Minute Walk Test [34]: primarily employed to evaluate aerobic capacity and endurance.
- Digital scale for body fat & weight (TANITA): to contribute to determine the subjects' physical condition.
- WHODAS 2.0 [35]: a standardized assessment of health and disability measurement.
- SF-12 Health Questionnaire [36]: health-related quality of life questionnaires.
- Activity of Daily Living Questionnaire [37]: a valid and reliable scale for the functional evaluation of patients.

We have already recorded the 45 required healthy subjects, and 14 subjects with functional limitations, currently being in the process of recruiting the pending 31 ones.

E. Database Labeling

After recording the available subjects, a database is being generated, and it needs to be classified, categorized, and identified to align with previously established metrics, understood as parameters for measuring functionality. This categorization and image annotation process is referred to as "Labeling".

The labeling process is a specialized task requiring specific annotation applications and is conducted by OTs specialized in evaluating functional limitations. This time-consuming yet fundamental task is essential in the system's training phases from both technical and clinical perspectives. We selected the ELAN labeling application [38] due to its ability to label temporal sequences and its versatility in labeling schemes. Fig. 3 shows an example of a labeling session.

We created a comprehensive labeling structure to facilitate the annotation of all available metrics, hierarchically organizing them to reduce labeling time. The set of required metrics is

quantitative, but they incorporate the qualitative analysis of OTs, that can determine when, how, and how much each functional metric occurs. For instance, a subject may grasp an object, but determining whether the grasp is functional at that moment can only be established by the criteria of a clinical specialist. Therefore, without the OTs involved in this process, the labeled data would not have clinical relevance.

F. System Modules

While generating the specifications of our proposal for the evaluation of functional limitations, we carried out an intense review work on the technological alternatives that could generate the required quantitative metrics and qualitative clinical inferences.

Again in a strong interdisciplinary fashion, we refined the initial list of metrics and inferences, generating a final list of 40 items to be estimated. For each item, we decided which specific data to generate (e.g., number, value, time, location, related object, hand involvement, sensor involvement, etc.) and which detection and estimation module could contribute in their estimation. The list of 40 items is too long to be included here, but the interested reader can access it in an online document (with their corresponding IDs).

Fig. 4 provides a general overview of the initial architecture of the system, including the sensor information sources and location (top); a general overview of the relationship between the selected detection/estimation modules, and the metrics/inference IDs (middle), and the processing modules and clinical output information to be generated (bottom), that considers clinical scores, and both motor and cognitive alterations.

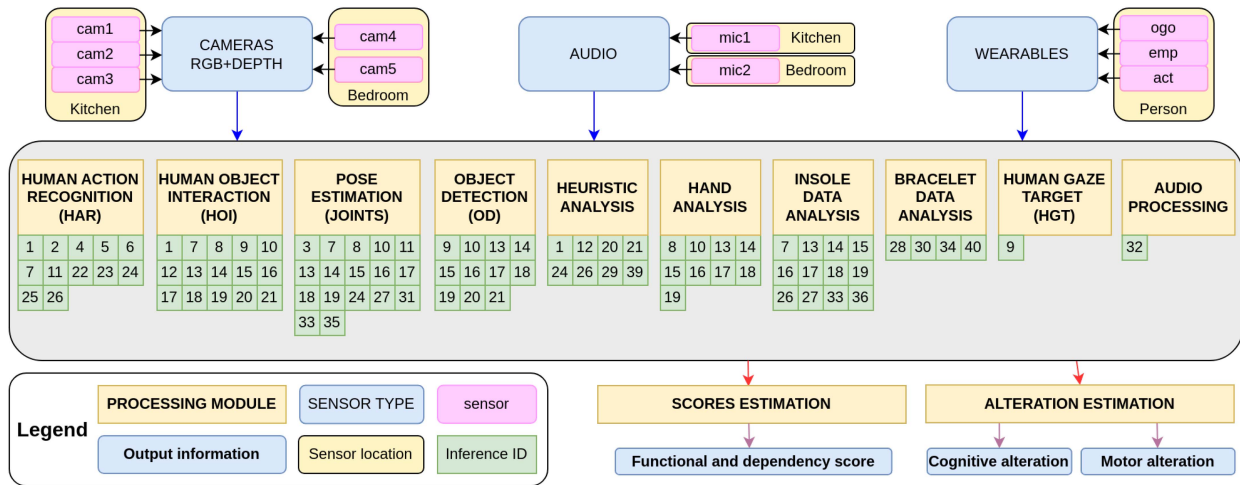


Fig. 4. General system architecture including sensor information, sensor location, processing modules and inference IDs.

In the following section we include an overview of the functionality of some of the detection and estimation modules, along with some quantitative and qualitative results with the progress achieved so far, and references to our current work.

IV. CURRENT STATUS OF SELECTED MODULES

A. Human and Object Detection

According to the project objectives, several objects have to be identified and located in the evaluation environment (kitchen and bedroom of the adapted apartment). These detected objects are later required by other modules such as Human Activity Recognition (HAR), Human-Object Interaction (HOI), or Human Gaze Target Estimation (HGT). Thus, this module [39] must detect and locate the different objects of interest, related to the analyzed ADLs [17], and classify them.

Several alternatives exist in the literature for object detection, but pre-trained models often struggle with identifying specific objects, especially smaller or deformable ones. In this work, we present two approaches depending on the nature of the objects: one uses neural networks for detecting rigid objects, while the other relies on color-based methods for deformable ones.

- *Rigid objects' detection:* we propose to use a set of models based on YOLOv8 [40]. Specifically, the larger objects ("person", "chair", and "table") are detected using the pre-trained COCO [41] weights, while for the other objects we use a one-object one-model approach, trained with the GEFAD-Objects database [39]. Furthermore, for the latter models, we use the SAHI inference method [42] in the EYEFUL environment. Additionally, we trained a head detector with the HollywoodHeads dataset [43], which is crucial for the HGT module. With this approach, we obtain the following metrics (for each object we show the precision and average precision, calculated at an Intersection Over Union factor (IoU) of 50% (AP50)): "plate" [0.7392, 0.7406]; "cutlery" [0.7564, 0.6960]; "shoes" [0.7793, 0.7465], "glass" [0.7314, 0.7105] and "head" [0.9531, 0.9570].

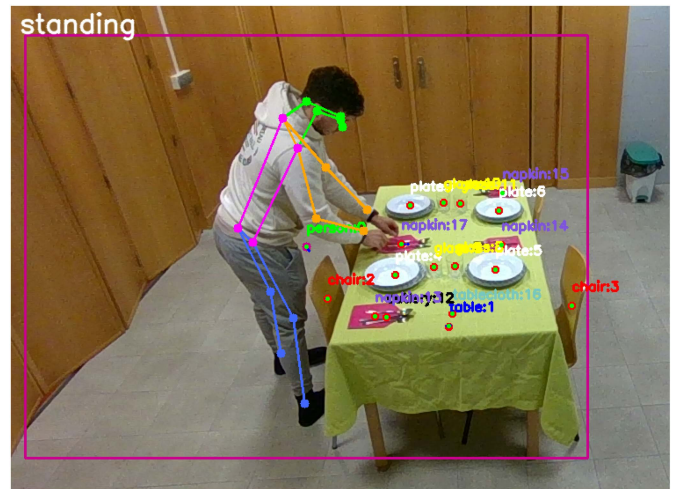


Fig. 5. Example of HAR and object detection with tracker.

- *Deformable objects' detection:* The second approach involves recognizing objects based on their color. This is particularly relevant for objects that are challenging to differentiate based on shape or size characteristics, such as the napkins, tablecloth, or socks. In this approach, the HSV color space is employed to segment and identify objects based on their color properties, and their sizes in the image. This allows segmenting the deformable objects, obtaining their position and size.

The human and object detection module is completed with a tracking algorithm based on a set of Kalman filters. In a first stage, a Kalman filter is created for each object, filtering their detections and estimating them when missing. Then, in a second stage, each object is tracked, setting an ID for their identification. The tracking module allows improving detections even if there are occlusions, and uses the tracking information for the HOI module. Fig. 5 shows an example of the OD+tracking module results.

TABLE I
BEST QUANTITATIVE RESULTS WITH GEFAD-HAR DATASET

Model	Pre-train	# frames	Batch	LR scheduler	Top1
I3D	Kinetics	32	6	Plateau	92.79
	SSv2	32	6	Plateau	89.75
TSM	Kinetics	8	10	MultiStep	86.40
	SSv2	8	10	Cosine	87.05
PoseC3D	NTU120	48	8	Cosine	89.21

B. Human Activity Recognition and Stress Detection

Human activity recognition (HAR) is also a relevant research area in applications in intelligent spaces. Given the specificities of our experimental setup, we carried out a data selection process from widely used HAR datasets (NTU [44], Moments in Time [45], and HMDB57 [46]). Five main actions have been considered as the most likely to occur during functional evaluations (standing, sitting, squatting, walking, and falling).

The finally selected HAR data subset (referred to as GEFAD-HAR) is composed of over 8,000 clips which were used to evaluate the HAR module with two of the main state-of-the-art methods for this purpose, depending on the input:

- *RGB data method:* In which two architectures have been evaluated, a 3D CNN (I3D) [47] and a 2D CNN (TSM) [48], based on ResNet-50. Both of them are pre-trained, trained and tested consecutively to determine the capacity of the collected datasets and the models used. For the training process, the weights from Kinetics [49] and Something-Something (SSv2) [50] were employed, and fine-tuning was performed to adjust the model weights for our specific application.
- *Skeleton data method:* For this second method, initial processing is required to obtain the person detection and acquire the skeleton joints, which will be used as input data for the HAR task. Fast R-CNN [51] and HRNet [52] models have been used for these processes, respectively. Subsequently, a pre-trained PoseC3D (SlowFast) [53] network is trained using the NTU120 dataset [44], doing fine-tuning with the data collected in the GEFAD-HAR video dataset.

Both methods achieve satisfactory results in the HAR task, as can be seen in Table I. However, using skeleton data as input yields better results due to the characteristics of our apartment environment. This is because the networks in the first method focus on local image features, which can lead to incorrect detection, while the second method uses the position and correlation of joints, aiding in detecting the chosen actions. Fig. 5 shows an example of a *standing* activity detection.

To prevent potential false detections, we designed a filter that prioritizes the action duration and confidence levels in order to determine whether the detection is accepted. Currently, a new algorithm is being developed to synchronize the results obtained from different perspectives of the scene.

We currently aim to also use the Empatica E4 wristband in the HAR task (training with the Opportunity dataset [54]) and also to provide information on the subjects' stress levels (given its negative impact in physical and mental health). Our stress detector adopts a minimally invasive approach and has

Looking at: napkin (36.65%), cutlery (33.55%), cutlery (29.8%)

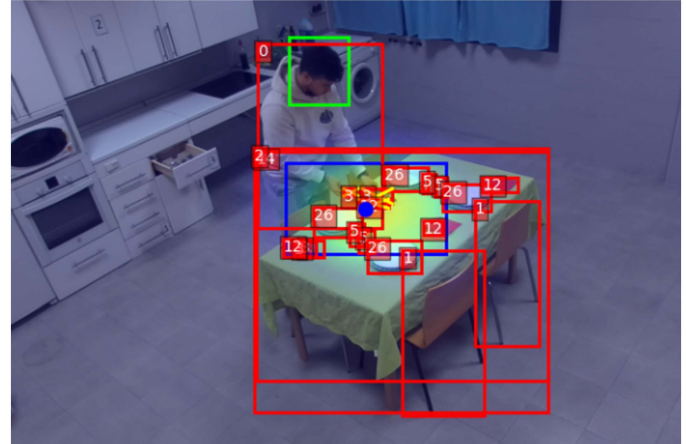


Fig. 6. Example of HGT and object detection.

been evaluated on the WESAD dataset [55]. We derive up to 295 features from the Empatica E4 wristband (temperature, electrodermal activity, and blood volume pulse), using methods such as the Fourier Synchrosqueezed Transform (FSST). This feature set is further reduced using the Minimum Redundancy – Maximum Relevance (MRMR) strategy, that are then used by several machine learning models. Our best results achieve a F_1 score of 91.72% outperforming the original WESAD study by 7.61%.

C. Human Gaze Target Estimation for Evaluation of Hand-Eye Coordination

One of the critical parameters for OTs when performing human functional assessment is evaluating hand-eye coordination, specifically determining whether the user looks at an object before interacting with it. Human Gaze estimation (GE) is a topic widely addressed in the state-of-the-art, but as stated in previous works [56], [57], [58], GE models only focus on estimating the gaze point, and are not interested in detecting the object looked at, which can have substantial practical utility. As far as we know, few works consider the Human Gaze Target Estimation (HGT) task [56], [58], [59], [60], [61], [62], and all of them with certain limitations. Therefore, we developed a method to infer the object being looked at, which combines these modules:

- *Gaze Estimator:* We used the model from [61], based on ResNet18 [63] as the feature extractor and an LSTM model [64] to exploit the temporal information and generate a more accurate output. The model was trained with the VideoAttentionTarget dataset [61], which is one of the most used benchmarks in the GE problem. This dataset is composed of videos from different TV shows, where the head bounding box (BB) and the respective gaze target have been labeled. This model generates a *heatmap*, that is shown in Fig. 6 as a yellow-ish blob.
- *Gaze orientation:* We are currently developing a new proposal to improve the results of [62]. To achieve that, we have trained a ResNet18 [63] with the Gaze360 dataset [65] to estimate the head's gaze orientation, using

as input the previously detected head BB. The features generated from this module will be later used to estimate the gaze point and the object looked at. At the moment, we report an average error of 17.6 degrees using the AdamW optimizer with 0.0005 as the initial learning rate, which is an acceptable error taking into consideration that we are using a light model and no temporal information.

After inferring the heatmap and the object information, we evaluate if the person is looking at an object following an improved version of [56]. Our strategy exploits the calculation of each object BB heatmap energy value to pre-select those with highest values. We then generate a *looked-at likelihood value* that depends on the IoU between the object BB and an estimated Region Of Interest BB (blue box in Fig. 6); and, finally, a probability of each object being looked at using a *softmax* operation.

Fig. 6 shows an example of an HGT result in which the system is able to propose the three most probable detections: looking at a napkin, and at two cutlery items next to it (see top of the figure for probability information).

D. Human Object Interaction (HOI)

The process by which humans interact with their physical environment, manipulating objects with their hands and upper limbs, is called manual manipulation. Analyzing this action has significant applications in motor assessment, requiring precise hand segmentation, three-dimensional pose estimation, grip type detection, and intention prediction.

Our hand segmentation algorithm is based on the ContactHands system [66] given its high accuracy and that it also provides the interaction type (with an object, with another person, with the person itself, or none). We have enhanced this solution by adding a pre-processing layer inspired by the SAHI inference approach [42], using a patch method centered on the detected person. We leverage the results from the person detection task (via a YOLOv8 [40] module) to reduce the network's search space, removing extraneous information that could hinder the hand detection task. Our initial results in the ContactHands's database, show an AP50 of 0.79, slightly improving the state-of-the-art results. Fig. 7 also shows how this approach (top row) improves the original results (bottom row) in our database. Fig. 8 show two additional examples of the hand segmentation module using the YOLO+ContactHands strategy.

Our current work in this task is oriented to combine the pose estimation (human body joints) with the depth information provided by the RGBD cameras to further improve the results.

E. Human Pose Estimation (HPE)

The estimation of the human body joints play a fundamental role in the determination of physical limitations, and the the person dexterity while performing ADLs. The Zed 2i cameras provide a reasonably good estimation of the body joints, and we are also evaluating state-of-the-art proposals, such as [67], which has reported to have high precision and efficiency.

Our initial efforts were oriented to the ETRI-Activity3D database [68], which contains recordings of young and elderly



Fig. 7. Comparison between ContactHands (bottom row) and YOLO+ContactHands (upper row). Detected hands are segmented in yellow.



Fig. 8. Hand segmentation examples.

individuals performing ADLs in three rooms of an apartment, in conditions very similar to those found in the environment of the EYEFUL project. In initial experiments with a single activity in the ETRI dataset (retrieving food from a fridge, which is similar to the EYEFUL action of grabbing objects from the kitchen cupboard), we achieved a Percentage of Detected Joints (PDJ, with an error lower than 10 cm) of 71.28% and an average joint position error of 4.33 cm.

We are currently evaluating other state-of-the-art alternatives that allow exploiting the availability of several synchronized cameras by the use of multi-view networks. Some of the studies we are currently evaluating are AdaFuse [69] and the proposal of Isakov et al. [70], which report the best literature results on the Human3.6 M database [71].

F. Human Balance Scoring

One of the factors that indicates low skill in individuals is balance, an element that has not been extensively analyzed in the execution of ADLs. For this purpose, some studies use IMUs [72] or RGBD cameras [73], [74].

The evaluation of a patient's balance during the performance of ADLs must be conducted at various phases. This is because,

while executing a task, there are moments of static balance and others of dynamic balance. The ultimate system we are striving to develop will use the human pose estimation (HPE) of the user to autonomously calculate an objective balance score.

As a previous study, first published in [19], we presented an approach to automate the assessment of balance in individuals that where standing still during the execution of an ADL. The proposed method involves estimating the person's Center of Mass (CoM) projection to the ground using theoretical and deep learning techniques from the 3D body joints position.

Our initial experiments on estimating the person's center of mass from the body joints resulted in a Mean Absolute Error (MAE) of 0.35 cm on a publicly available database provided in [75]. We also proposed neural models to learn clinically validated objective balance scores, from the estimated center of mass and center of pressure. These proposals improved the state-of-the-art in the Sensory Orientation (subscale within the BESTest scale) estimation task [76], achieving an error of 0.611 points on the dataset.

Our current work is devoted to estimate an objective balance score that is valid for both static and dynamic balance conditions, to exploit it in the functional assessment task. The proposal will combine information from the body joint estimations, and that coming from the pressure sensors in the Moticon OpenGo insoles.

G. Scores and Alteration Estimation Modules

These modules will be in charge of generating clinical scores and information on the possible cognitive and/or motor alterations that can be associated to the values of the estimated metrics and inferences. Our initial strategies in these modules are inspired by the Rasch analysis [77], and we will also exploit machine learning approaches.

To assess our progress toward the goal of these modules, we conducted a series of initial experiments using some of the metrics and inferences included in Fig. 4, in order to initially evaluate their discrimination capabilities. These experiments were framed as a classification task with two classes (control subjects and those with functional limitations). We used a rigorous leave-one-out cross validation methodology (training with all subjects but one, evaluating with the one left out, and averaging the performance results). Given the clear class imbalance, we evaluated the system not only on overall accuracy, but also on metrics sensitive to class imbalance. Our best experiment so far using the XGBoost classifier [78] got an accuracy of 91.3%, F_1 score of 88.0%, Precision of 89.1% and Recall of 87.1%, that are very promising considering the experimental conditions and that we have not yet carried out extensive experimental work in this line.

While this classification task is only one component of our project, it is a relevant milestone within the broader project objectives. Ultimately, our work on the scores and alteration estimation modules is ongoing, but these initial classification results provide a strong foundation, and we are confident that the final system will perform well in these advanced tasks.

H. Clinical Validation Design

The EYEFUL project will also carry out a thorough clinical validation process. Both ethical and significance considerations have been taken into account during the dataset design and recording processes. The selection of well established scales that are administered to all the subjects, guarantees that the clinical validation will assess the validity of the system to estimate accurate values in the functional assessment task.

The validation will comprise both a content validation (consisting of the analysis and selection of activities and parameters), and the clinical validation per se, that will evaluate whether it identifies, measures or acceptably predicts clinical, biological and physical aspects in the indicated context (including a specific population).

Validation of specific pathologies is not within the project's scope, but it is planned for future initiatives of the research groups in charge of the EYEFUL project.

V. CONCLUSIONS AND FUTURE WORK

In this paper, we have presented a proposal that aims to represent a significant step toward automating the assessment of human functional limitations, underscoring the importance of interdisciplinary collaboration and advanced technology. The project's holistic approach considers various components of functional assessment, including cognition, motor skills, and executive functions. Within the EYEFUL project, we have created an intelligent apartment, equipped with state-of-the-art technology that facilitates relevant data collection from subjects while carrying out ADLs. This rich dataset, which is being properly labeled, will play a crucial role in developing and rigorously testing fusion algorithms and data analysis methodologies. The participation of healthcare professionals closely working with the engineering team guarantees the focus on the clinical validity of the results.

We have provided details on the full user-centric methodology, design and deployment processes, from the selection of ADLs, the description of the carefully selected hardware equipment (sensors, processing and communication devices), and the database generation and labeling processes, including subject recruitment criteria (90 subjects in two balanced cohorts of people with and without functional limitations).

We also described an overview of the design of the machine learning systems aimed at generating all the metrics and inferences required by the healthcare professionals involved in the project. We have also presented the current status of some of the main technological modules, including quantitative and qualitative results obtained to date.

In conclusion, this research presents pioneering work in the automation of functional assessments. By highlighting the interdisciplinary collaboration, the innovative deployment of intelligent environments, and the promising initial results, we underscore the potential impact and applications of our system in the field of healthcare.

Our ongoing and future work focus on expanding our dataset to include a more diverse cohort, enhancing the robustness of our machine learning models. We also aim to refine our existing

modules, by incorporating more sophisticated algorithms and improving their accuracy and reliability. Comprehensive clinical validations will be conducted to ensure the system's clinical relevance and utility, involving both content and clinical validation. Additionally, we plan to explore applications of our system in specific neurological pathologies. Tailoring our approach to these conditions will further contribute to healthcare technology and improve the quality of life for individuals with functional impairments. We will continue our close collaboration with healthcare professionals to ensure that our system meets the highest standards of clinical relevance and utility.

It is worth noting that one of the limitations of the proposal is the potential invasion of people's privacy when using sensors such as cameras and microphones, which should be considered for future implementations. There are several approaches to ensure people's privacy [79], from alternatives that use data that does not allow people to be identified, (such as depth data [80], [81] or human joints [53]), to strategies related to Privacy Preserving Machine Learning (PPML), using techniques like Federated Learning [82] that avoid sharing private data, or Secure Multi-Party Computation (MPC) [83].

ACKNOWLEDGMENT

We gratefully acknowledge the computer resources at Artemisa, funded by the European Union ERDF and Comunitat Valenciana as well as the technical support provided by the Instituto de Física Corpuscular, IFIC (CSIC-UV).

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